Oct. 30, 1962 P. CURETON 3,061,003

VERTICAL BLIND

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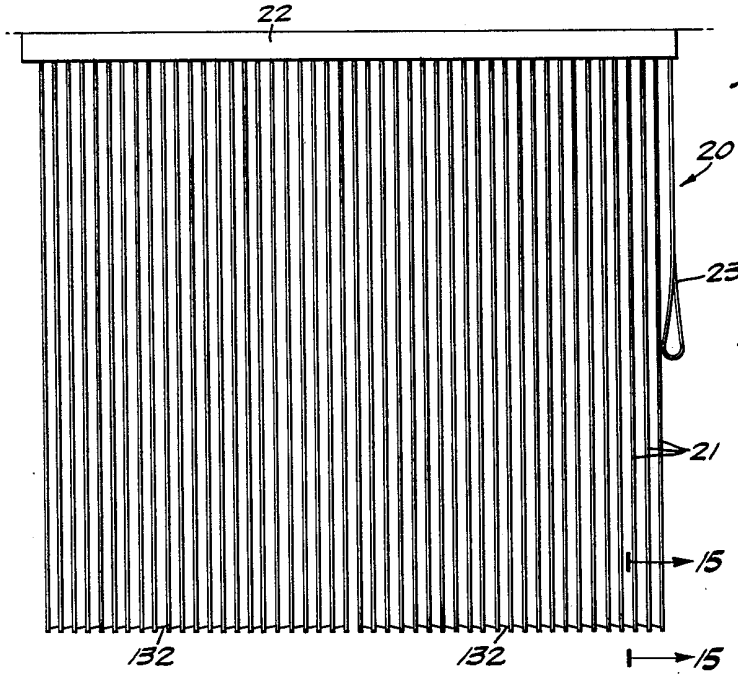


FIG. 1.

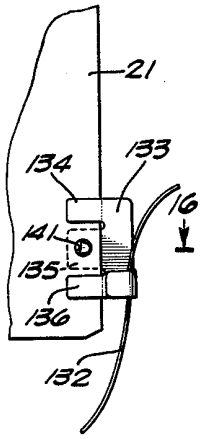


FIG. 15.

FIG. 2.

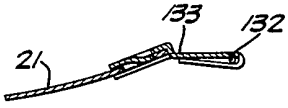


FIG. 16.

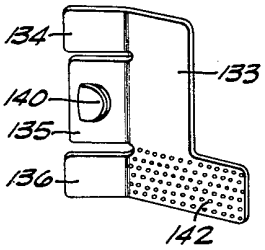
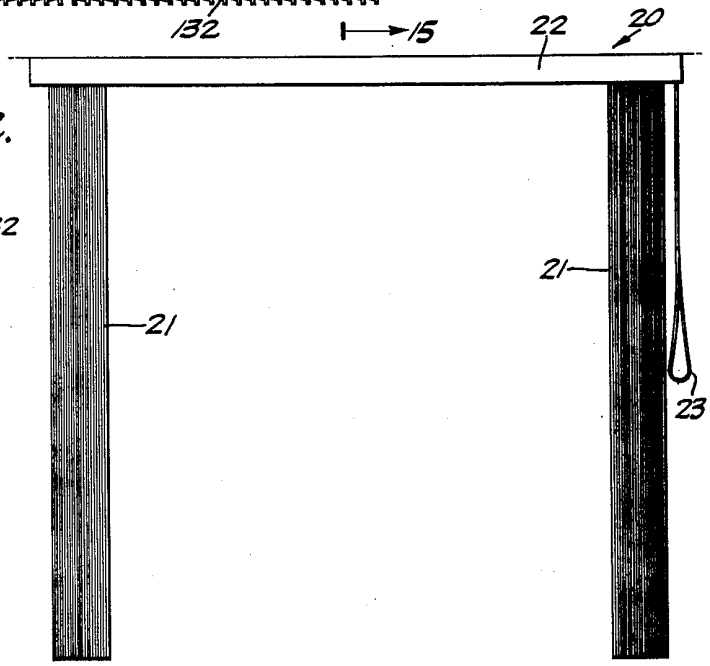


FIG. 17.



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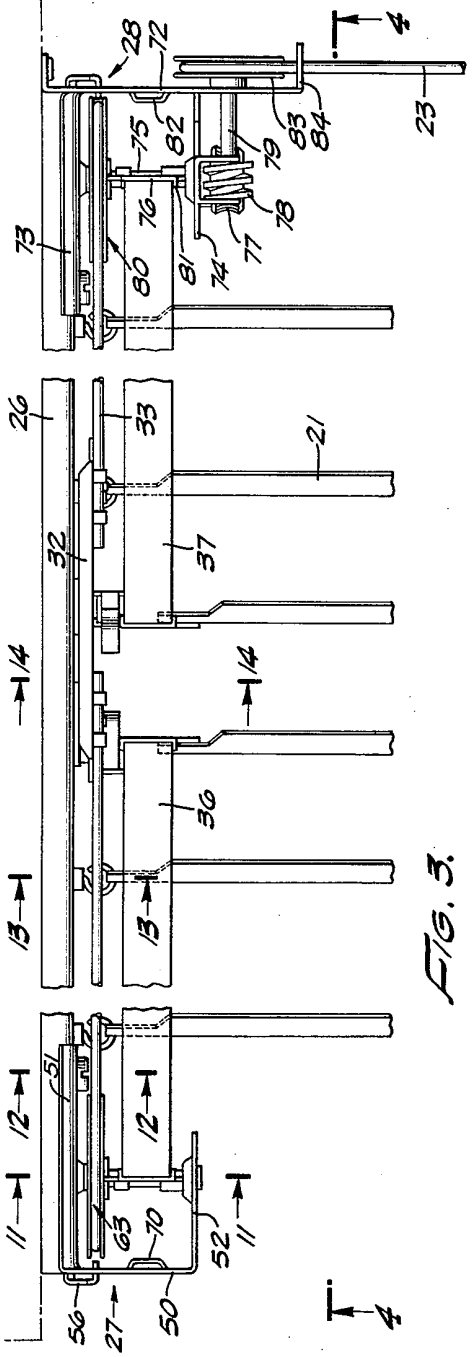


FIG. 3.

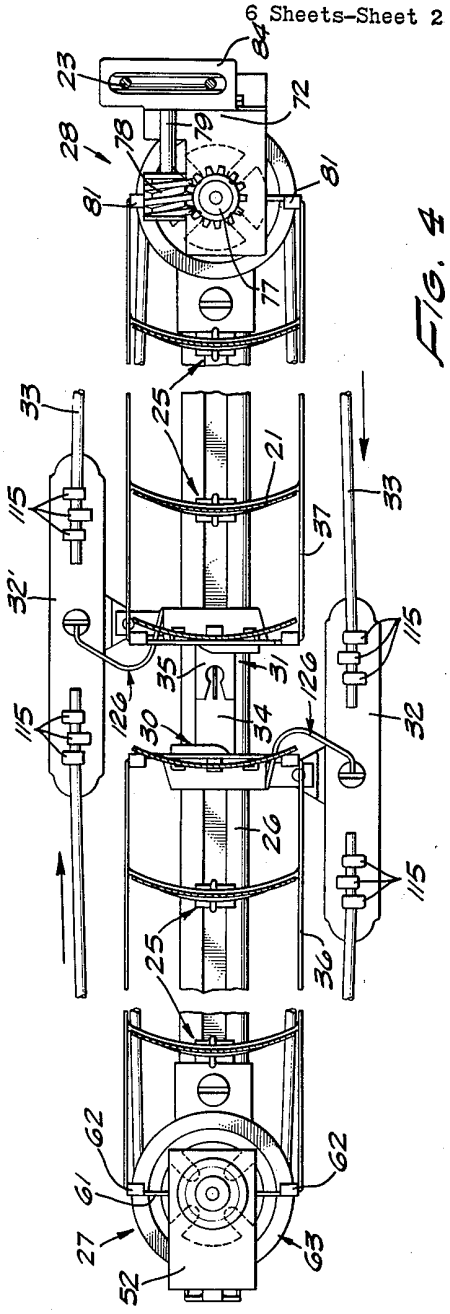


FIG. 4

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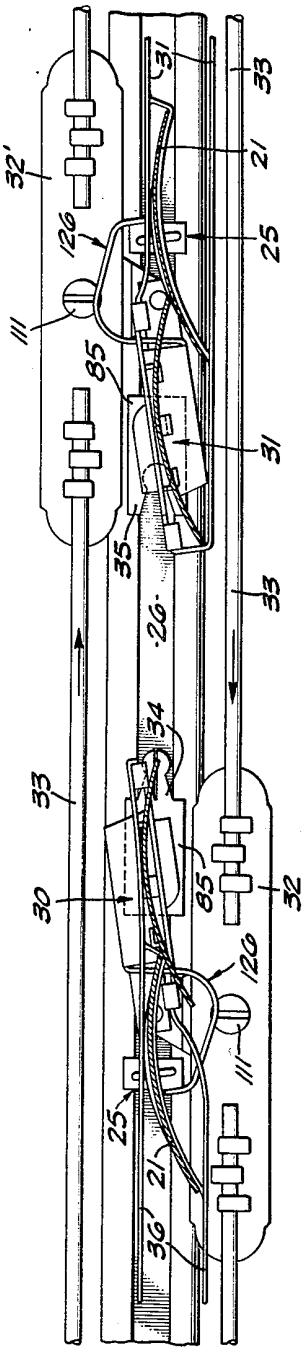


FIG. 5.

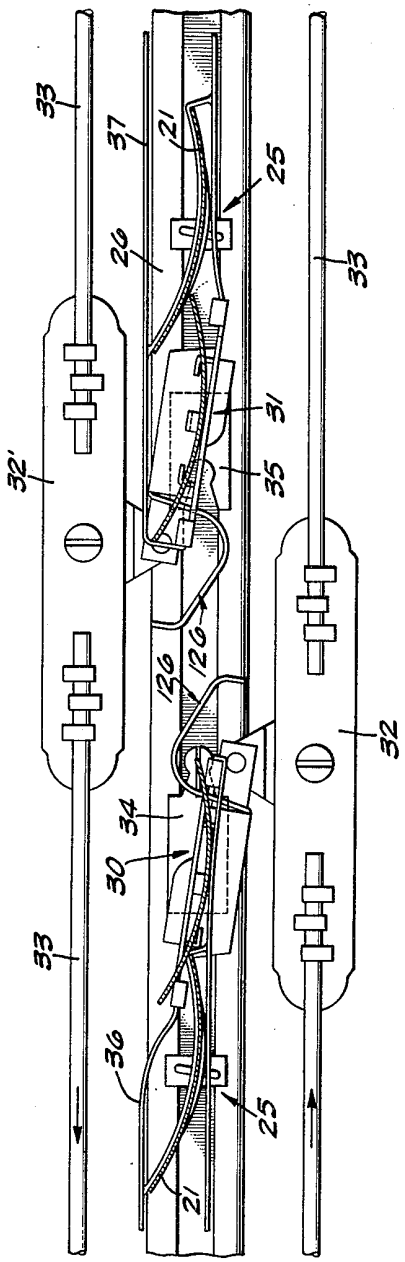


FIG. 6.

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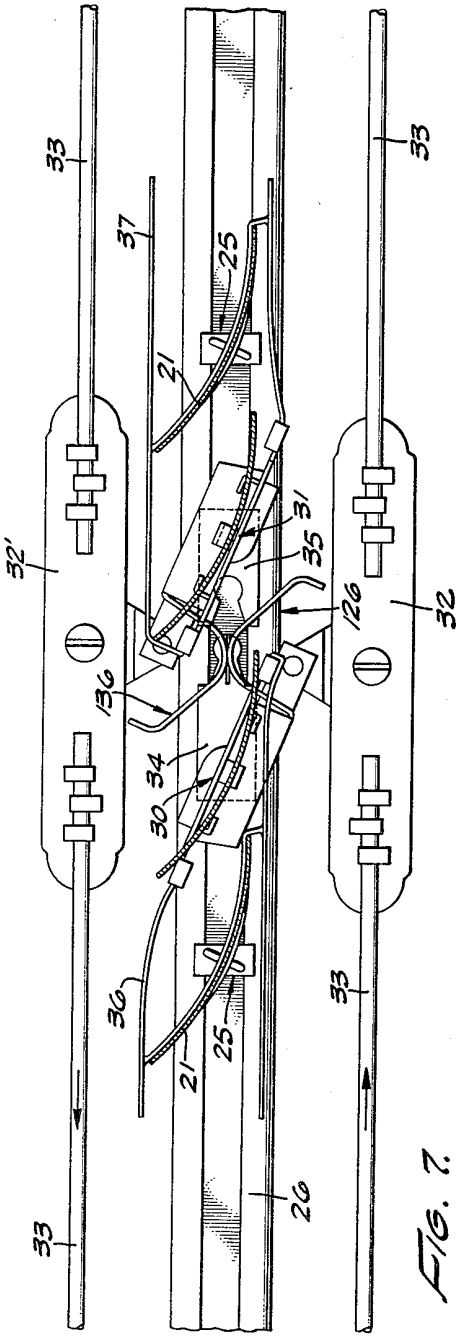


FIG. 7.

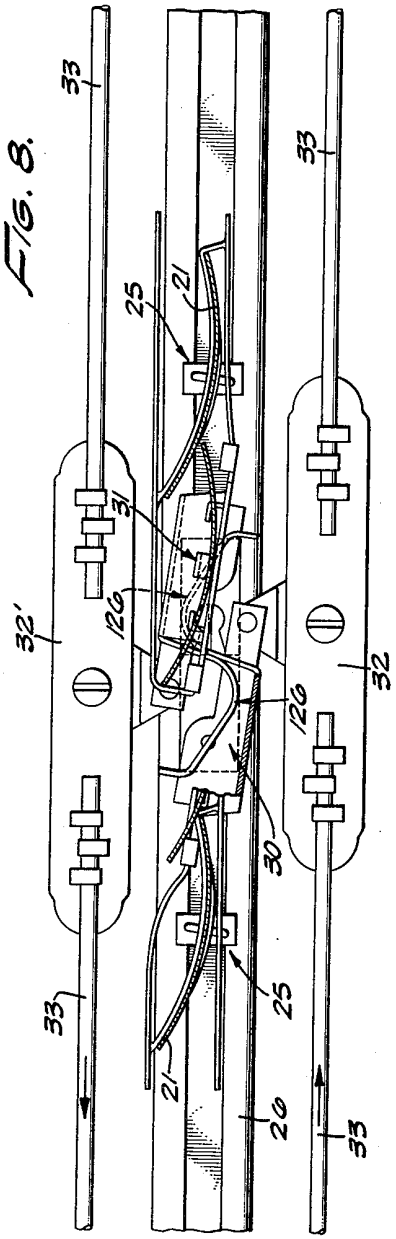


FIG. 8.

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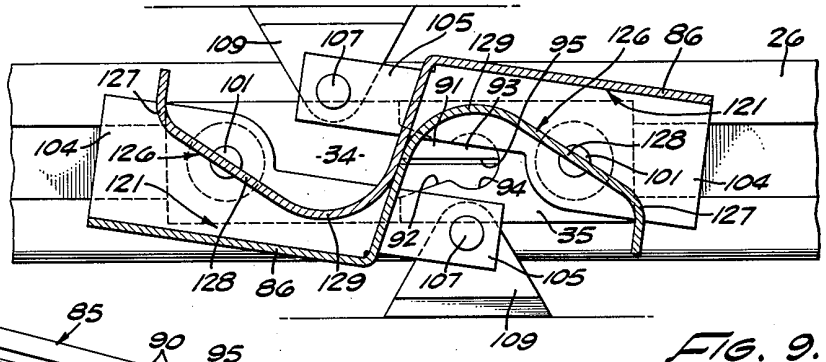


FIG. 9.

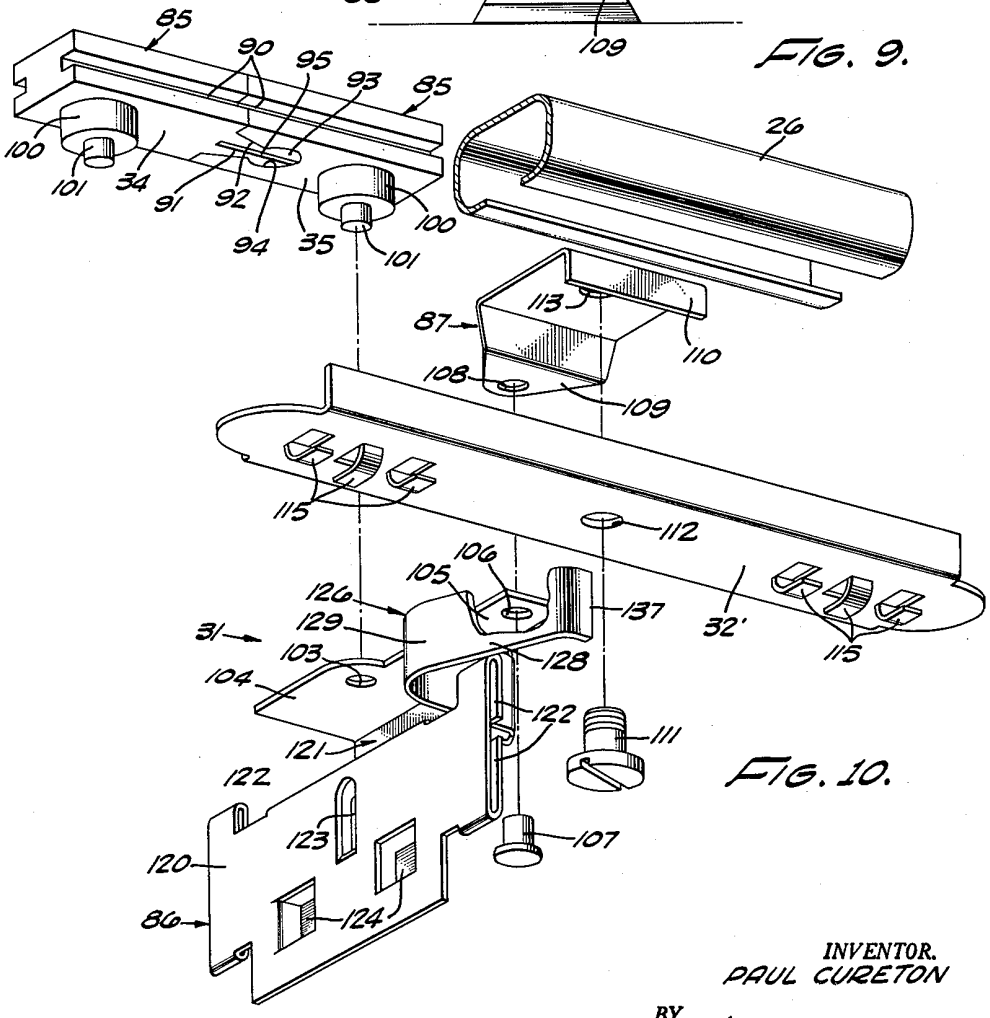


FIG. 10.

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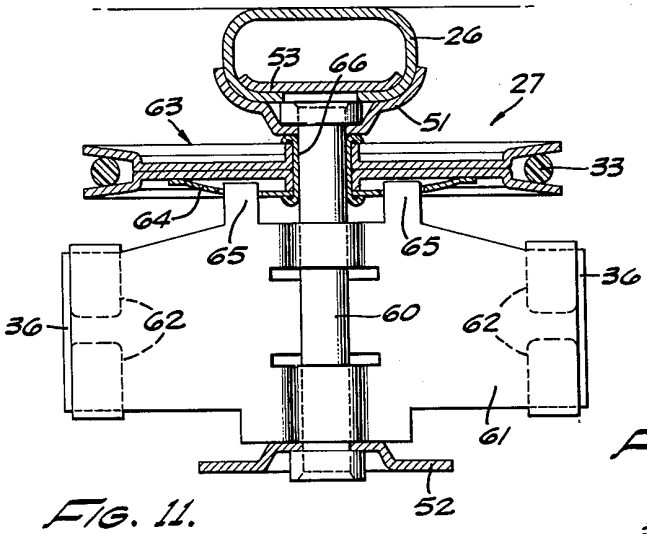


FIG. 11.

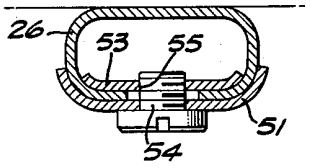


FIG. 12.

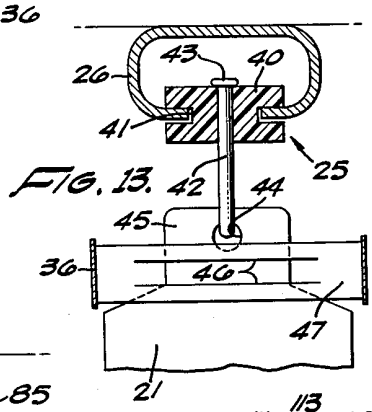


FIG. 13.

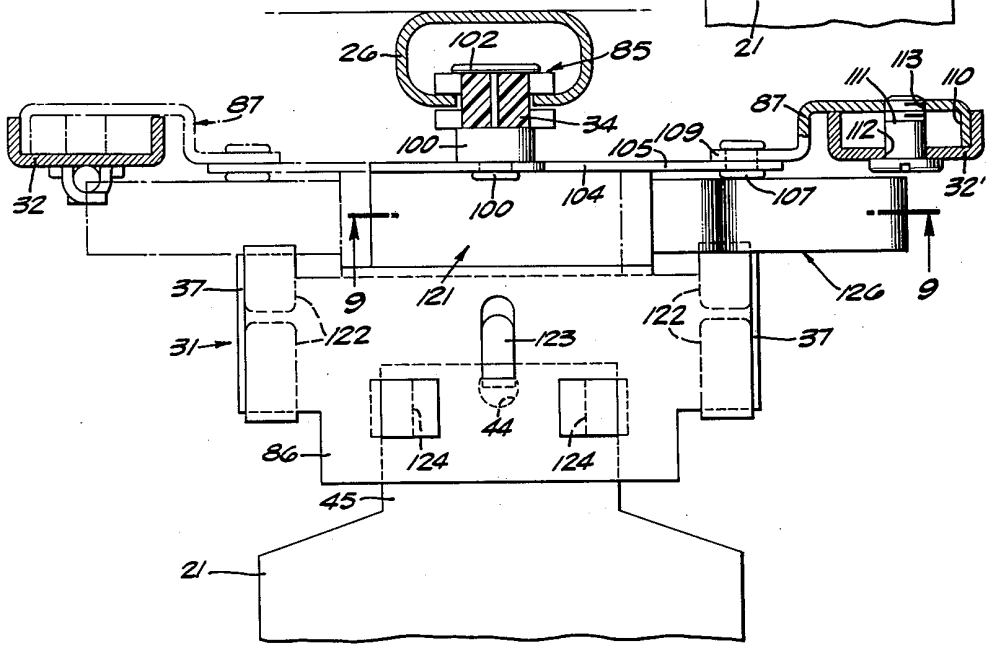


FIG. 14.

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United States Patent Office

3,061,003
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3,061,003

VERTICAL BLIND

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(450 Springs Road, Vallejo, Calif.)

Filed Sept. 30, 1955, Ser. No. 537,674
8 Claims. (Cl. 160—118)

The present invention relates to vertical slat blinds, and its primary object is to provide a new and improved blind wherein the slats are drawn between open and closed positions, and also tilted through substantially 180° about their longitudinal axes, by a single control cord.

Another object of the invention is to provide a blind of the class described wherein the slats come together to form compact bundles when drawn to the sides of the window.

A further object is to provide a vertical slat blind that is relatively simple and inexpensive to manufacture and assemble, while still another object is to provide a blind wherein the slats close tightly, giving complete, shadowproof privacy.

The foregoing and other objects and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the preferred embodiment thereof, reference being had to the accompanying drawings, wherein:

FIGURE 1 is an elevational view of a vertical slat blind embodying the principles of the invention, showing the blind in the fully extended condition;

FIGURE 2 is a similar view of the blind, showing the slats drawn to the sides thereof;

FIGURE 3 is an enlarged fragmentary elevation of the top ends of the slats and the operating mechanism therefor;

FIGURE 4 is another view of the same, looking upward into the mechanism at 4—4 in FIGURE 3;

FIGURE 5 is a view similar to FIGURE 4, but showing the center slat carriers and attached linkage being drawn apart;

FIGURE 6 is a similar view, showing the slat carriers and linkage being drawn together;

FIGURE 7 shows the center slat carriers approaching one another, and the cam arrangement that separates the two adjacent center slats so that they come together in the proper manner;

FIGURE 8 shows the center slat carriers fully closed together and interlocked, so that the slats may be tilted without being drawn apart;

FIGURE 9 is an enlarged horizontal section through the cams of the two center slat carriers, taken at 9—9 in FIGURE 14;

FIGURE 10 is an exploded view of the locking sliders and one of the associated slat carriers;

FIGURE 11 is an enlarged sectional view, taken at 11—11 in FIGURE 3, through the idler pulley assembly at one end of the blind;

FIGURE 12 is an enlarged sectional view, taken at 12—12 in FIGURE 3;

FIGURE 13 is an enlarged sectional view taken at 13—13 in FIGURE 3;

FIGURE 14 is an enlarged vertical section taken at 14—14 in FIGURE 3;

FIGURE 15 is an enlarged fragmentary view of one of the clips that hold the cord tying the slats together at their bottom ends;

FIGURE 16 is a sectional view taken at 16—16 in FIGURE 15; and

FIGURE 17 is a perspective view of the clip in FIGURE 15, before the tab has been bent over to secure the cord.

In the drawings, the blind of the invention is designated in its entirety by reference numeral 20, and is seen to comprise a plurality of vertically hanging slats 21, which are supported on operating mechanism concealed behind a fascia board 22. The slats are controlled by an endless cord 23, which functions both to draw the slats to the sides, as shown in FIGURE 2, and also to tilt the slats about their longitudinal axes, as will be described in more detail presently. The slats 21 are suspended from carriers 25 which are slidably mounted on a standard drapery traverse track 26 that is attached to a suitable support, such as a cornice board or the like. At one end of the track 26 is an idler assembly 27, and at the other end thereof is a tilter assembly 28.

The blind illustrated herein is of the center pull type, divided into two equal groups of slats which are adapted to be drawn to both sides of the window opening in the manner of draperies, as shown in FIGURE 2, or drawn together at the center to cover the window opening from side to side, as shown in FIGURE 1. The center slats of the two groups are suspended from master carrier assemblies 30 and 31, which are attached by cable bars 32 and 32' to a traverse cable 33. The master carrier assemblies include co-acting latch members 34 and 35 that interlock when the center slats are drawn together, to prevent separation of the two groups of slats as the slats are tilted. Connected between the idler assembly 27 and the corresponding master carrier assembly 30 is a ladder tape 36; while a second ladder tape 37 is connected between the tilter assembly 28 and its corresponding master carrier assembly 31.

Each of the slat carriers 25 comprises a slider 40, preferably in the form of a block of nylon or other pastic, having grooves 41 formed in the opposite sides thereof, which receive the confronting edges of the traverse track 26. Extending vertically through a hole in the center of the block 40 is a hook 42, which is headed at 43 to retain the hook on the block. The hook 42 is inserted through a hole 44 in the top end of the slat 21, and thus supports the latter. Each of the slats 21 is cut back on both side edges at the top end thereof to form a tab 45 of reduced width. The tab 45 is inserted through slits 46 in the rungs 47 of the ladder tapes 36, 37, and the slats are thus tied together by the tapes and are rotated in unison thereby.

The idler assembly 27, which is shown at the left-hand end of the traverse track in FIGURES 3 and 4, and further illustrated in detail in FIGURES 11 and 12, comprises a sheet metal bracket 50, having horizontally projecting top and bottom legs 51 and 52. The top leg 51 is shaped to fit against the underneath side of the traverse track 26, and the edges thereof are turned up to follow the contour of the track. Projecting into the inside of the track 26 is a clamping plate 53, which is shaped to lie flat against the bottom flanges of the track. A clamping screw 54 passes through a hole in the top leg 51 of the bracket and is screwed into a threaded hole 55 in plate 53. When the screw 54 is tightened up, plate 53 is drawn toward the leg 51, clamping the edges of the track 26 between them. The projecting end of the plate 53 is bent downwardly at 56 and hooked into the bracket 50, as shown in FIGURE 3. Extending through aligned apertures in the top and bottom legs 51 and 52 of the bracket is a vertical shaft 60, which is headed at both ends to lock it against endwise movement with respect to the bracket. A paddle, or blade 61, is rotatably mounted on the lower portion of the shaft 60, and at the outer ends of the paddle are tabs 62, which are bent down over the ends of the ladder tape 36 to clamp the latter to the paddle.

Fixed on the top end of the shaft 60 is a pulley 63, which is frictionally connected to the paddle 61 by means of a friction disc 64 that engages upstanding ears 65 on

the paddle. The pulley **63** comprises two sheet metal stampings which are arranged back to back, as shown in FIGURE 11, and joined together by a tubular hub member **66**. The top and bottom ends of the hub member **66** are rolled outwardly to engage annular flanges on the two pulley halves. The friction disc **64** is concentric with the pulley **63**, and is confined between the rolled flange at the bottom end of the hub member **66** and the associated annular flange on the lower half of the pulley. The friction disc **64** is dished slightly, and its outer peripheral edge bears against the underneath side of the pulley for frictional contact therewith. The traverse cable **33** passes around the pulley **63**, and the latter is turned on its shaft **60** when the cable is moved in one direction or the other. Due to the frictional driving connection between the pulley **63** and paddle **61** afforded by the friction disc **64** and ear **65**, the paddle is turned with the pulley until one end or the other of the paddle strikes a stop **70** (see FIGURE 3) on the bracket **50**. At this point, the paddle is prevented from turning any further, but the pulley is free to continue rotating on the shaft **60**, and merely slides on the friction disc **64**.

The tilter assembly **28** at the right hand end of the traverse track **26** is likewise comprised of a vertical bracket plate **72** having inwardly bent upper and lower legs **73** and **74**. The upper leg **73** is shaped to lie against the underneath side of the traverse track **26**, and is clamped to the track by means of a clamping plate similar to plate **53** of the idler assembly **27**. A vertical shaft **75** extends between the upper and lower legs **73**, **74** of the bracket, and rotatably mounted on the mid-portion of the shaft is a paddle, or blade **76**, similar to the paddle **61**. A pulley **80** is fixedly mounted on the upper end of the shaft **75** and frictionally connected to the paddle **76** by a friction disc, in the same manner that the pulley **63** of the idler assembly is connected to the paddle **61**. Tabs **81** at the ends of the paddle **76** are bent downwardly over the ends of the ladder tape **37** to clamp the latter to the paddle. Thus, the pulley **80** has a frictional driving connection with the paddle **76**, and turns the latter until one end of the paddle engages a limit stop **82** on the bracket. The shaft **75** projects below the leg **74** of the bracket, and mounted on the bottom end thereof is a pinion **77**, which is driven by a worm **78**. The worm **78** is fixedly mounted on a horizontal shaft **79**, which is journaled on the bracket **72**, and fixed to the other end thereof is a pulley **83**, over which the control cord **23** is trained. The cord **23** is confined to the pulley **83** by a guide flange **84** projecting laterally from the bottom end of bracket **72**; the said flange having a closed slot through which the cord **23** passes.

The two master carrier assemblies **30** and **31** are identical except for the latch members **34**, **35**. The master carrier assemblies are best shown in FIGURES 9, 10 and 14, and it will be noted that each assembly comprises a slider **85**, bracket **86**, and link **87**. The sliders **85** are preferably molded of nylon, and are in the form of elongated blocks having grooves **90** in opposite sides thereof to receive the edges of the traverse track **26**. The male latch member **34** has a tapered post **91** projecting longitudinally from one end of the slider block, and the female latch member **35** has a correspondingly shaped notch **92** formed in its adjacent end to receive the post **91**. The post is provided with a rounded enlargement **93** at its outer end, and the notch **92** has a similar enlarged opening **94** to receive the enlargement **93** on the post. The post **91** is slit longitudinally at **95** to permit the side thereof to squeeze together as the enlargement **93** is forced through the narrow portion of the notch **92** just ahead of the opening **94**. As the enlargement **93** moves all the way into the opening **94**, the two sides of the post spring apart, and the male latch member **34** is resiliently interlocked with the female latch member **35**.

Projecting downwardly from the bottom side of the slider **85** is a cylindrical stud **100** having a central aperture to receive a pin **101**. The pin **101** extends upwardly through the slider **85** and is headed at **102**, as shown in FIGURE 14. The downwardly projecting end of the pin **101** extends through a hole **103** in a horizontal flange **104** of the bracket **86**, and the bottom end of the pin is upset to rivet the bracket to the slider. A lateral projection **105** of the flange **104** is apertured at **106** to receive a rivet **107**, which also passes upwardly through a hole **108** in an ear **109** of the link **87**. The link **87** passes up over the adjacent edge of the channel-shaped cable bar **32'**, and has a downwardly turned flange **110** which abuts against the bottom of the cable bar. A screw **111** passes upwardly through a hole **112** in the cable bar, and is screwed into a tapped hole **113** in the link **87**. At each end of the two cable bars **32**, **32'**, are groups of three downwardly struck tabs **115**, which are spaced apart from one another along the longitudinal axis of the cable bar. The tabs **115** face alternately in opposite directions, and the traverse cable **33** is threaded under the tabs, which are then pinched down on the cable to fasten the latter securely to the cable bar.

The bracket **86** is bent downwardly from the flange **104**, then horizontally for a short distance back under the flange, and finally downwardly again to form a paddle **120**. The configuration resulting from the above-described bends provides a recessed portion **121**, the purpose of which will soon become apparent. Tabs **122** at the ends of the paddle **120** are bent down over the ends of the associated ladder tape **36**, **37**, to secure the tapes to the paddle. Struck out from the paddle **120** at the center thereof is an upwardly extending hook **123** which is adapted to be inserted through the hole **44** in the center slat to support the latter. Inwardly projecting tabs **124** are struck out from the paddle on either side of the hook **123**, and engage the edges of the slat tab **45** to hold the slat parallel to the paddle.

Projecting from one end of the recessed portion **121** of the bracket plate **86** is a cam strip **126** comprising a rounded leading edge **127**, inclined ramp **128**, and rounded heel **129**. The cam strips of the two master carriers **30**, **31**, ride up over one another as the carriers come together at the center of the blind, and their purpose is to rock the bracket plates **86** slightly so that the leading edges of the center slats are tilted away from one another momentarily to prevent either one of the slats from getting behind the other. As the rounded heel portions **129** of the cam strip pass completely over one another, each of them drops into the recessed portion **121** of the other bracket plate, and the two bracket plates resume their original angular position. While the two cam strips are thus interlocked, as shown in FIGURE 9, the angle of contact between them is substantially transverse to the longitudinal axis of the traverse track; hence, they tend to cam the latch members **34**, **35** together during the final interlocking movement of the latch members.

The bottom ends of the slats **21** are tied together in properly spaced relation by means of a string **132**, which is attached to the slats by slat links **133**, as best seen in FIGURES 15, 16 and 17. Each of the slat links **133** is provided with three vertically spaced tabs **134**, **135** and **136** along one edge thereof, which are adapted to pass on opposite sides of the slat. The center tab **135** has a tongue **140** struck out to one side thereof, which drops into and engages the edge of a hole **141** in the slat, thereby securing the link to the slat. Another tab **142** is provided on the opposite edge of the link which is bent over onto the string **132** to pinch the latter so that the link is attached to the string. The inner surface of the tab **142** and the portion of the link which it overlies are roughened, as shown in FIGURE 17, to prevent the string from sliding through the link.

The operation of my invention is as follows: With the blind fully closed, as shown in FIGURES 3 and 4, the slats are tilted by pulling on the control cord **23**, which operates the tilter **28**. The pulley **80** of the tilter is rotated in one direction, carrying the traverse cable

33 with it. This moves the two cable bars 32, 32' in opposite directions, which causes the two master carrier bracket plates 86 to rotate in unison. The traverse cable 33 also rotates the idler pulley 63, which turns the paddle 61 in unison with the bracket plate 86 of the master carrier assembly 30. The two ladder tapes, 36, 37, are thus tilted simultaneously by their respective master carrier assemblies and paddles 61, 76, which causes all of the slats 21 to rotate in unison.

To draw the slats to the sides of the window, the cord 23 is pulley in the direction to cause the traverse cord 33 to travel in the direction of the arrows shown in FIGURE 4. The slats are first tilted in the clockwise direction to the limiting position, and then the latch members 34, 35 are pulled apart. Continuing pull on the cord 23 draws the two master carriers 30, 31 toward their respective ends of the traverse track 26, pushing the other slat carriers 25 ahead of them as they go. FIGURE 5 shows the master carriers 30, 31 being drawn apart, as just described.

To draw the slats together, the cord 23 is pulled in the opposite direction, which first rotates the paddles 61, 76 and bracket plates 86 through approximately 180 degrees, after which the master carriers 30, 31 are pulled toward the center of the traverse track by the cable 33. FIGURE 6 shows the master carriers approaching one another at the center of the blind. As the slats approach still closer (see FIGURE 7) the cam strips 126 engage and ride up over one another, rocking the bracket plates 86 in the clockwise direction, so that the leading edges of the center slats are momentarily turned away from one another to prevent the slats from getting behind one another. During the final increment of their travel, the latch members 34, 35 lock together, as shown in FIGURE 8, at which point the slats can be tilted open by a reverse pull on control cord 23.

While I have shown and described in considerable detail what I believe to be the preferred form of my invention, it will be understood that various changes may be made in the shape and arrangement of the several parts without departing from the broad scope of the invention, as defined in the following claims.

I claim:

1. A vertical slat blind comprising a traverse track, a plurality of carriers slidably supported on said track, slats depending from said carriers, flexible ladder means having transverse rungs attached to said slats to space said slats apart uniformly along the length of said track and to rotate said slats in unison, a pair of vertically disposed shafts rotatably supported at opposite ends of said track, a pulley fixedly secured to each of said shafts, a paddle journaled on each of said shafts and frictionally driven by said pulley, each of said paddles having its ends connected to the ends of said flexible ladder means, a limit stop limiting the rotation of said paddle in either direction, an endless cord trained around said pulleys and extending parallel to said track, said cord being attached to an end carrier at a point spaced radially from the axis of rotation of the attached slat, whereby lengthwise movement of said cord causes said end carrier to rotate to one extreme position of angularity and thereafter to slide along said track, and means for driving one of said pulleys to move said cord in either direction to operate said slats.

2. A vertical slat blind comprising a traverse track, a plurality of carriers slidably supported on said track, slats depending from said carriers, said slats being divided into two groups which are adapted to be drawn together at the center of said track, or drawn back to opposite ends thereof, flexible ladder means for each of said groups of slats, said ladder means having transverse rungs attached to said slats to space the slats apart uniformly along the length of said track and to rotate said slats in unison, each of said groups of slats having a master carrier at the inner end thereof, pivoted means

at each end of said track, each of said ladder means being attached at one end thereof to one of said pivoted means and at the other end to its associated master carrier, whereby all of the slats in each group are tilted in unison by said master carriers or are shifted thereby between extended and retracted positions, said master carriers having coacting latch members which interlock to yieldingly hold said master carriers together at the center of said track, a rotatable pulley at one end of said track, and a cord trained around said pulley and extending along both sides of said track, one length of said cord being attached to one of said master carriers and the other length thereof being attached to the other master carrier, said cord being attached to said master carriers at points spaced radially from the axis of rotation of the attached slat, whereby lengthwise movement of said cord causes said master carriers to turn in unison to one extreme position of angularity and thereafter to slide in opposite directions along said track, and means for moving said cord in either direction, said cord being operable either to tilt said slats or to draw the same between extended and retracted positions.

3. A vertical slat blind comprising a traverse track, a plurality of carriers slidably supported on said track, slats depending from said carriers, said slats being divided into two groups which are adapted to be drawn together at the center of said track, or drawn back to opposite ends thereof, flexible ladder means for each of said groups of slats, said ladder means having transverse rungs attached to said slats to space the slats apart uniformly along the length of said track and to rotate said slats in unison, each of said groups of slats having a master carrier at the inner end thereof, said master carriers having coacting latch members which interlock to yieldingly hold said master carriers together at the center of said track, a pair of vertical shafts rotatably supported at opposite ends of said track, a pulley fixedly secured to each of said shafts, an endless cord trained around said pulleys and extending lengthwise of said track along opposite sides thereof, a paddle journaled on each of said shafts and frictionally driven by said pulley, said paddles being connected at their ends to the adjacent end of their respective ladder means, a limit stop for limiting the rotation of said paddles in either direction, one length of said cord being attached to one of said master carriers and the other length thereof being attached to the other master carrier, said cord being attached to said master carriers at points spaced radially from the axis of rotation of the attached slat, whereby lengthwise movement of said cord causes said master carriers to turn in unison to one extreme position of angularity and thereafter to slide in opposite directions along said track.

4. The invention set forth in claim 3, with the addition that said master carriers have cooperating cam means which engage and ride up over one another as the master carriers approach the center of said track, said cam means being operable to tilt the slats depending from said master carriers so that the leading edges of the slats are momentarily turned away from one another, thereby preventing the center slats from getting behind one another.

5. A vertical slat blind comprising a traverse track, a plurality of carriers slidably supported on said track, slats depending from said carriers, a pair of members disposed at opposite ends of said plurality of carriers and pivoted intermediate their ends for rotation in a horizontal plane, one of said members being stationary with respect to said traverse track and the other being slidably supported thereon, flexible ladder means having side tapes attached at their ends to the ends of said rotatable members, said side tapes being interconnected by transverse rungs which are attached to said slats to space the slats apart uniformly along the length of said track and to rotate said slats in unison, means for rotating said members so as to shift said side tapes in opposite directions and thereby tilt said slats, and a control cord ex-

tending lengthwise of said track on one side thereof and back on the other side, said cord being connected to said other member so that when said cord is moved in either direction, said other member and the associated slats are first tilted in one direction or the other and then are moved along said track in one direction or the other.

6. A vertical slat blind comprising a traverse track, a plurality of carriers slidably supported on said track, slats depending from said carriers, a pulley journaled for rotation about a vertical axis adjacent one end of said track, a paddle mounted for rotation coaxially with said pulley and frictionally driven thereby, a second paddle pivotally mounted on the carrier remote from said pulley, flexible ladder means having side tapes attached at their ends to the ends of said paddles, said side tapes being interconnected by transverse rungs which are attached to said slats to space the slats apart uniformly along the length of said track and to rotate said slats in unison, a cord passing around said pulley and connected to one end of said carrier-mounted paddle, and means for driving said pulley so as to rotate said paddles to one extreme position and thereafter pull said remote carrier along the length of said track to extend or retract said carriers and slats.

7. A vertical slat blind comprising a traverse track, a plurality of pivoted carriers slidably supported on said track, slats depending from said carriers, flexible ladder means having transverse rungs attached to said slats to space the slats apart uniformly along the length of said track and to rotate the slats in unison, pivoted means adjacent one end of said track, said ladder means being attached at one end thereof to said pivoted means, a cord extending lengthwise of said track from said one end thereof to the other end, means adjacent the other end of the track returning said cord to said one end, the end carrier remote from said one end of said track being attached at a point spaced radially from the pivot center thereof to said cord on one side of said track, whereby longitudinal movement of said cord causes said end carrier to rotate to one extreme position of angularity and thereafter to slide along said track, the other end of said ladder means being attached to said end carrier and movable thereby, whereby all of said slats are rotated simultaneously by said end carrier and are extended or retracted thereby, latch means on said end carrier engageable by companionate latch means on said other end of said track for yieldingly holding said end carrier at the fully extended position on said track, whereby said slats can be tilted without drawing said end carrier along the length of said track, and means for moving said cord in either direction.

8. A vertical slat blind comprising a traverse track, a plurality of carriers slidably supported on said track, slats depending from said carriers, a pair of pivoted members disposed at opposite ends of said plurality of carriers and pivoted intermediate their ends for rotation in a horizontal plane, one of said members being stationary with respect to said traverse track and the other being slidably supported thereon, flexible ladder means having side tapes attached at their ends to the ends of said pivoted members, said side tapes being interconnected by transverse rungs which are attached to said slats to space the slats apart uniformly along the length of said track and to rotate said slats in unison, operating means having a driving connection with one of said pivoted members whereby the latter may be turned in either direction through substantially 180° of rotation so as to shift said side tapes in opposite directions and thereby tilt said slats, and a control cord extending parallel to said track and connected to said other pivoted member, said operating means having a driving connection with said control cord whereby the latter is moved lengthwise of said track when said operating means is actuated, the movement of said control cord causing said other pivoted member and the associated slats to be rotated first in one direction or the other and then to be moved along said track in one direction or the other.

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